

Clinical and Oral Microbiology

The world of Microbes: Their Impact on Health

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Objectives

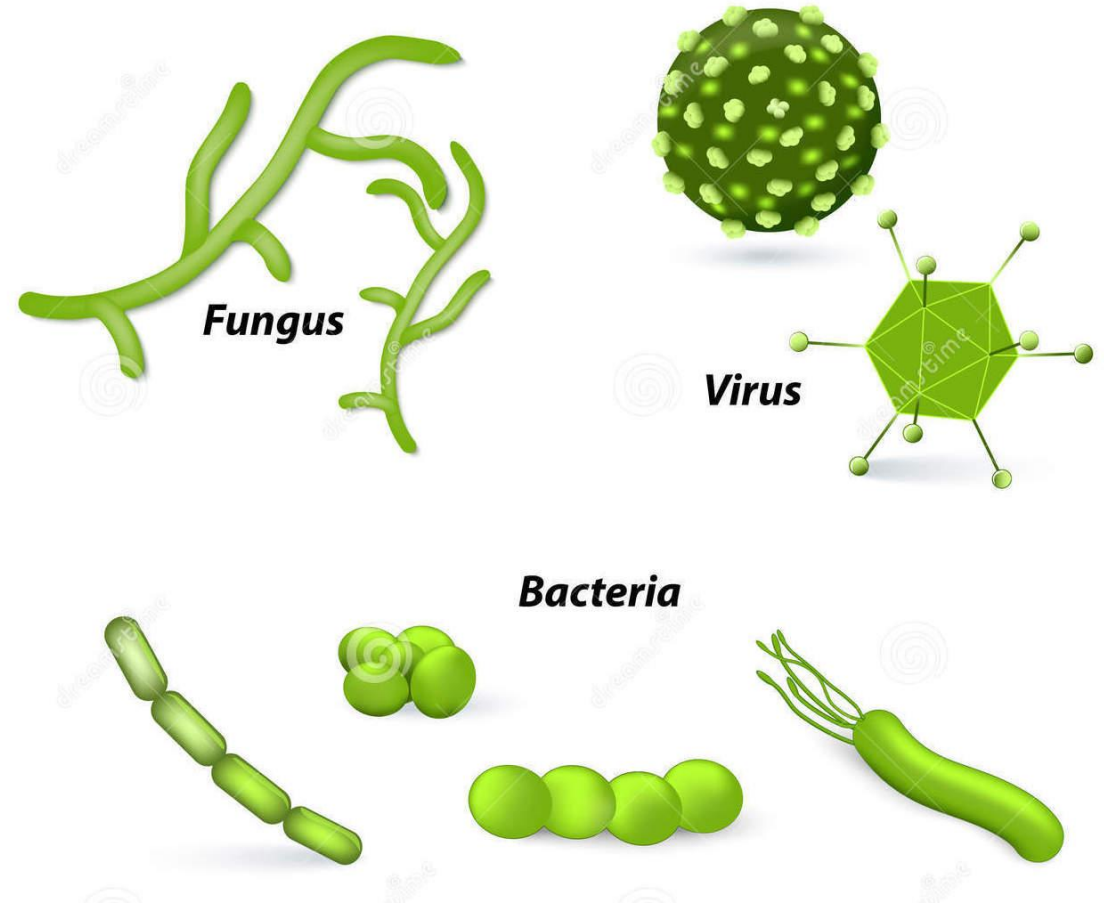
1. Describe the structure of a bacterial cell and its replication cycle.
2. Describe the function of different cellular structures.
3. Describe main features of bacterial virulence and pathogenicity.
4. Describe medical significance of select Gram-positive and – negative bacteria.

Outline

- What is a microbe?
- Bacterial anatomical structure
- Bacterial replication
- Bacterial pathogenicity
- Stages of pathogenesis
- Examples of pathogenic bacteria

What is a microbe?

- A microbe is microscopic organisms that -----(**can or can't**) be seen by naked eye.
- Different life forms including:
 - Bacteria
 - Viruses
 - Fungi
 - Parasites



Microbial life-forms

Prokaryotes “bacteria”

Always unicellular.

No nucleus or organelles

DNA is circular with no protein

Ribosome: small 70S (30S and 50S)

Cell division: binary fission

Reproduction: asexual

Eukaryotes “fungi & parasites”

Often multicellular.

Always have nucleus and organelles

DNA: linear with proteins

Ribosome: Large 80S (40S and 60S)

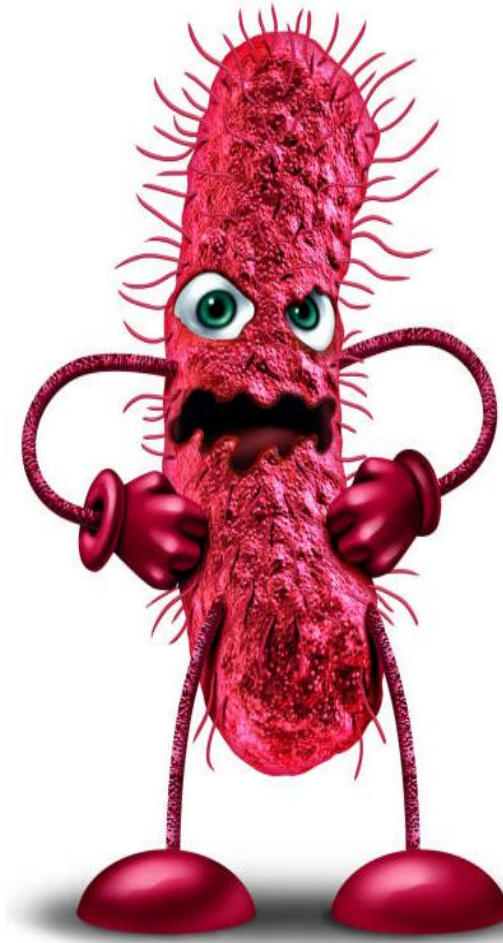
Cell division: Mitosis and meiosis

Reproduction: Sexual and asexual.

Bacteriology

Bacterial Family

- *Acidobacteriota* (from *Acidobacterium*)
- *Actinomycetota* (from *Actinomyces*)
- *Aquificota* (from *Aquifex*)
- *Armatimonadota* (from *Armatimonas*)
- *Atribacterota* (from *Atribacter*)
- *Bacillota* (from *Bacillus*)
- *Bacteroidota* (from *Bacteroides*)
- *Balneolota* (from *Balneola*)
- *Bdellovibrionota* (from *Bdellovibrio*)
- *Caldisericota* (from *Caldisericum*)
- *Calditrichota* (from *Caldithrix*)
- *Campylobacterota* (from *Campylobacter*)
- *Chlamydiota* (from *Chlamydia*)
- *Chlorobiota* (from *Chlorobium*)
- *Chloroflexota* (from *Chloroflexus*)
- *Chrysiogenota* (from *Chrysiogenes*)
- *Coprothermobacterota* (from *Coprothermobacter*)
- *Deferribacterota* (from *Deferribacter*)
- *Deinococcota* (from *Deinococcus*)
- *Dictyoglomota* (from *Dictyoglomus*)



- *Elusimicrobiota* (from *Elusimicrobium*)
- *Fibrobacterota* (from *Fibrobacterota*)
- *Fusobacteriota* (from *Fusobacterium*)
- *Gemmatimonadota* (from *Gemmatimonas*)
- *Ignavibacteriota* (from *Ignavibacterium*)
- *Kiritimatiellota* (from *Kiritimatiella*)
- *Lentisphaerota* (from *Lentisphaera*)
- *Mycoplasmata* (from *Mycoplasma*)
- *Myxococcota* (from *Myxococcus*)
- *Nitrospinota* (from *Nitrospina*)
- *Nitrospirota* (from *Nitrospira*)
- *Planctomycetota* (from *Planctomyces*)
- *Pseudomonadota* (from *Pseudomonas*)
- *Rhodothermota* (from *Rhodothermus*)
- *Spirochaetota* (from *Spirochaeta*)
- *Synergistota* (from *Synergistes*)
- *Thermodesulfobacteriota* (from *Thermodesulfobacterium*)
- *Thermomicrobiota* (from *Thermomicrobium*)
- *Thermotogota* (from *Thermotoga*)
- *Verrucomicrobiota* (from *Verrucomicrobium*)

&
more

Oral Cavity

The oral microbial community is a diverse microbial group that colonizes the human body.

Bacteria living on the human body or within the human body outnumber the total number of cells by about 10 to 1.

Gram-positive bacteria

Streptococcus mutans
S. sanguinis
S. oralis
S. mitis
S. gordonii
S. parasanguinis
S. salivarius
S. anginosus
Gemella morbillorum
Rothia dentocariosa
Actinomyces naeslundii
A. gerencseriae
A. odontolyticus
A. oris
Filifactor alocis
Lactobacillus salivarius
L. fermentum
L. plantarum
Bifidobacterium dentium
Eubacterium nodatum
Parvimonas micra
Peptostreptococcus anaerobius
Propionibacterium acnes

Gram-negative bacteria

Fusobacterium nucleatum
F. periodonticum
Haemophilus parainfluenzae
Porphyromonas gingivalis
P. endodontalis
Prevotella intermedia
P. loescheii
P. denticola
P. melaninogenica
P. nigrescens
Tannerella forsythia
Bacteroides odontolyticus
Neisseria subflava
Veillonella parvula
Aggregatibacter actinomycetemcomitans
Capnocytophaga ochracea
C. gingivalis
Campylobacter rectus
C. ureolyticus
Treponema denticola
T. socranskii
T. vincentii

Bacterial Structure

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graph TD; A[Bacterial Structure] --> B[Cell Envelop<br/>Outer structure]; A --> C[Appendages]; A --> D[Cell Interior]
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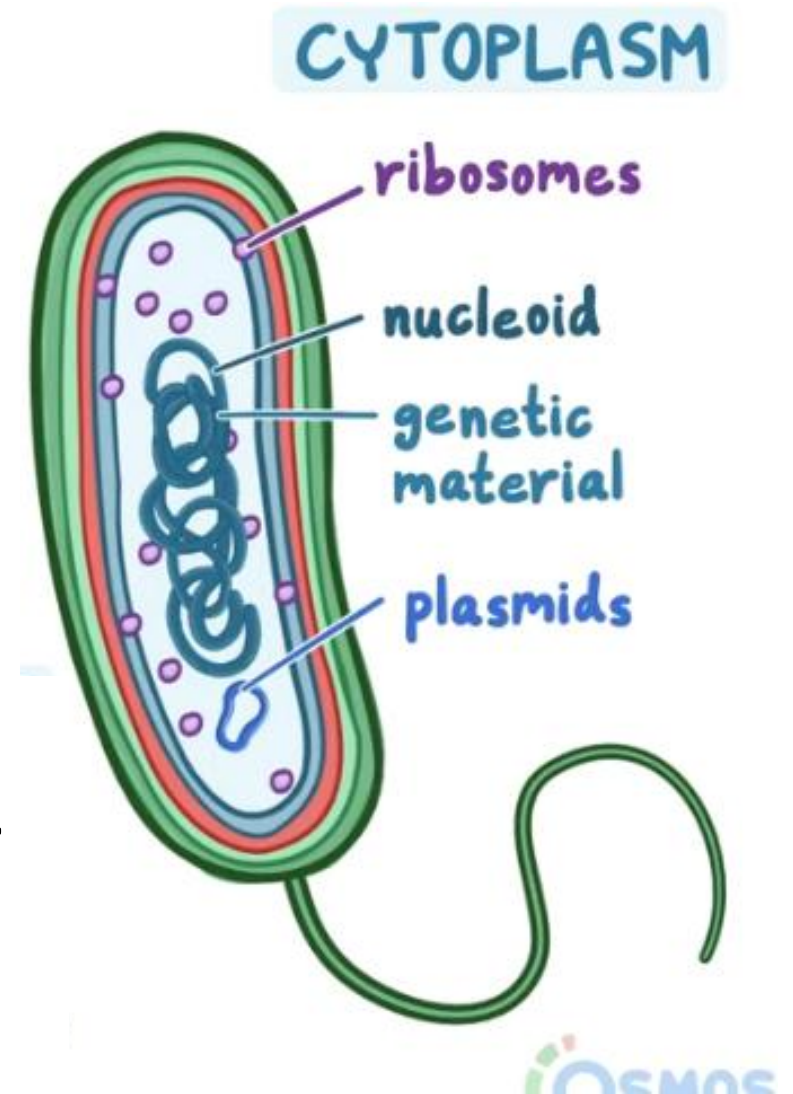
Cell Envelop
Outer structure

Appendages

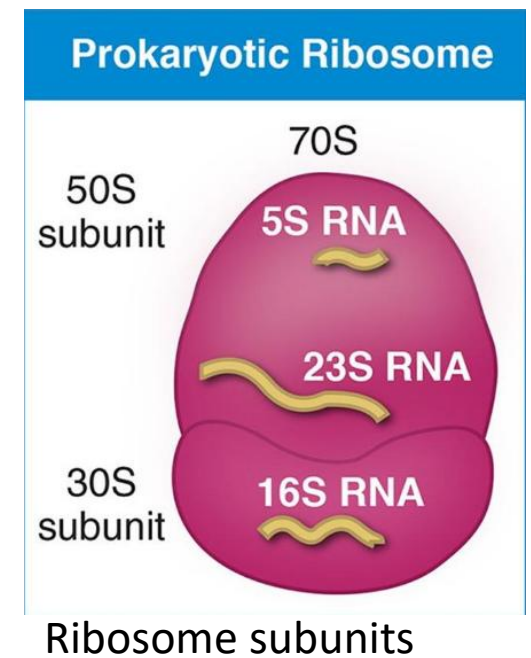
Cell Interior

Cell Interior

- No subcellular organelles or cellular compartments.
- **Bacterial cytoplasm** contains the ribosomes, double stranded genetic material (nucleoid) and plasmids.
- **Nucleoid:** circular, supercoiled DNA; small amount of RNA, not surrounded by a nuclear membrane



Ribosomes: contain proteins and RNAs. The bacterial ribosomes are composed of 30S (small subunit) and 50S (large) subunits (together, sedimentation coefficient of 70S).



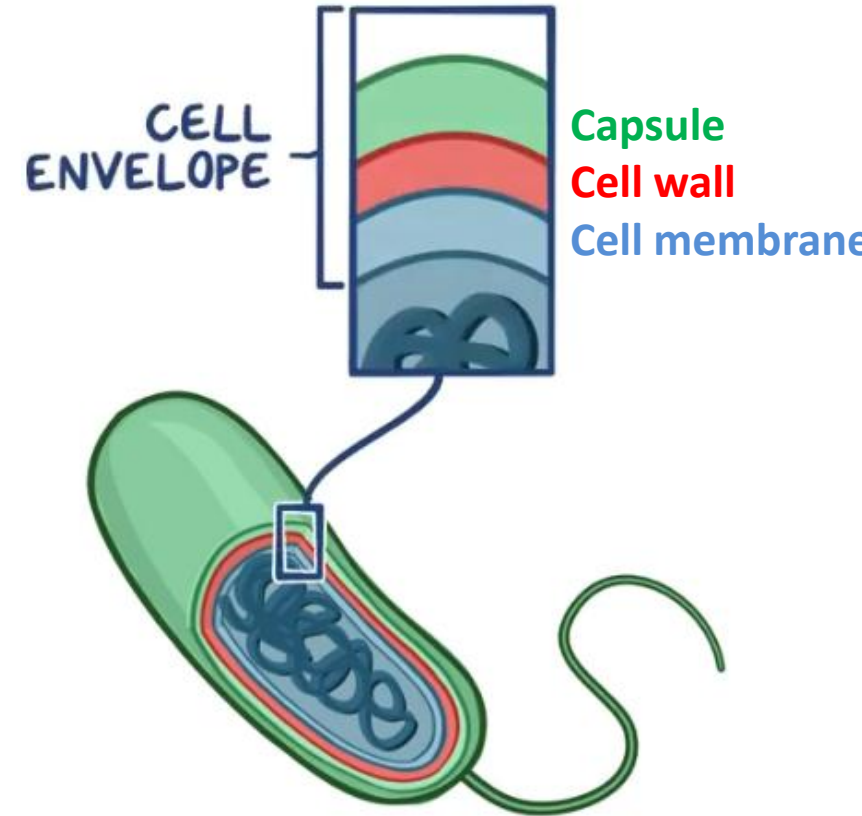
Plasmid: circular non-chromosomal double stranded DNA. Contains genes essential for bacterial protection. Can replicate independently and can transfer from one bact. to another. Can contain genes for antibiotic resistance or virulence factors. Research tool.

Transposons: small DNA pieces that move between bacterial DNA and plasmids. They can code for antibiotic resistance or bacterial toxins.

Cell Envelope

A) Cell membrane: phospholipid bilayer.

- Contains carrier lipids, **penicillin-binding proteins** (imp. for cell wall biosynthesis) and enzymes.
- Allows the active transport of nutrients.
- Site of action for certain antibiotics.
- Imp. for lipid synthesis, respiration and photosynthesis.



B) Cell wall: Confers Gram-staining characters, induce the host immune response, protection and maintain bacterial shape.

1- **Gram-positive bacteria:** the cell wall is composed:

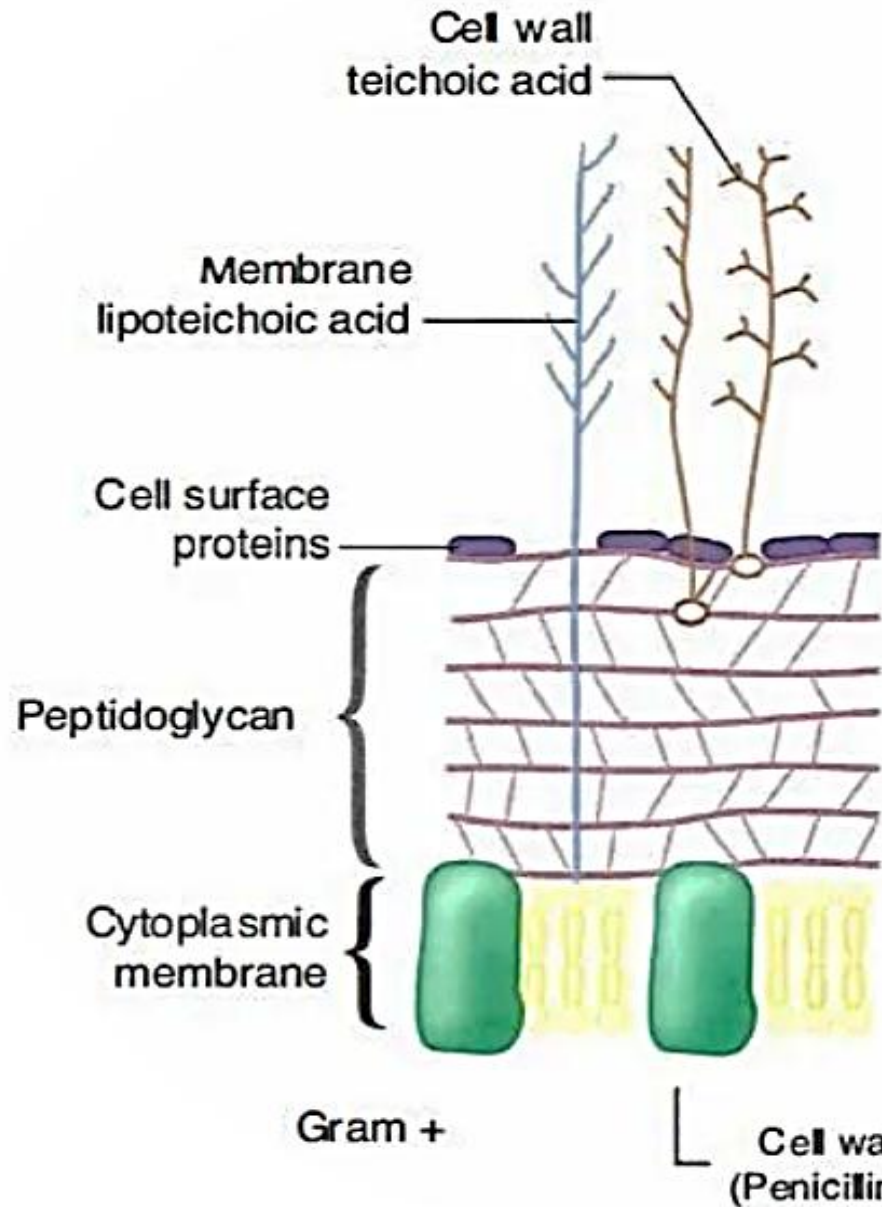
- Thick peptidoglycan layer.
- Lipoprotein and polysaccharide polymers (teichuronic acid, **Teichoic acids** or **lipoteichoic acids**).
- Lipoteichoic Acids and Teichoic acid are unique to G +ve. Contribute to the structural integrity of G +ve bacteria. Important in pathogenesis (adhesion to the host and avoidance of immune surveillance).

2- Gram-negative bacteria: Inner + outer membrane

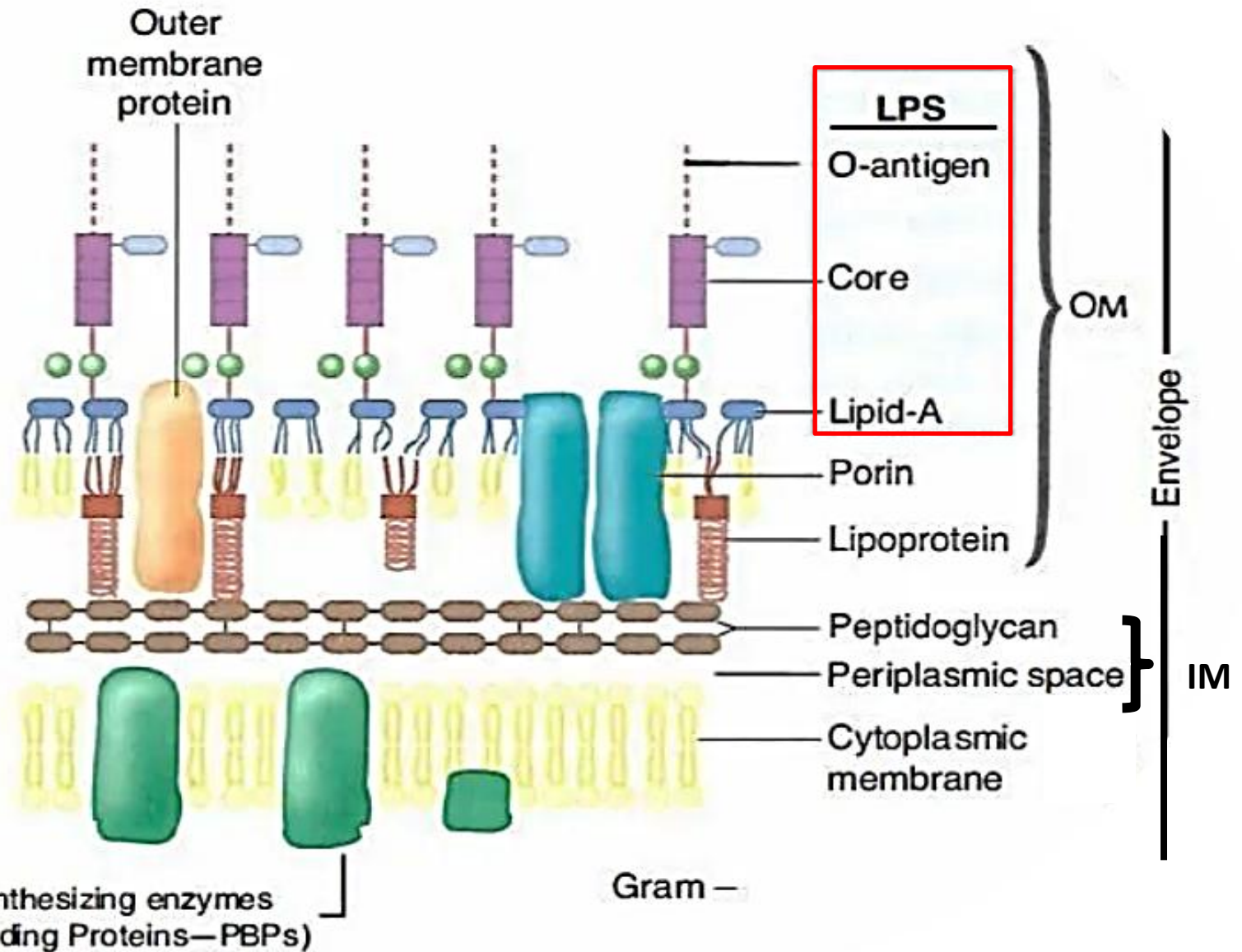
- Inner membrane: thin layer of peptidoglycan in the inner membrane.
- Outer membrane: lipopolysaccharide (LPS; aka endotoxin) and proteins. LPS, unique to G -ve, include:
 - O-antigen: target for antibodies.
 - Lipid A, toxic component, is released from dying bacteria and it triggers pathogenic symptoms (including fever, inflammatory response, apoptosis and septic shock) via inducing TNF and IL-1.

Certain species may contain **β -lactamase enzyme** between in the outer and inner membrane (several G -ve and *Staph.* “G+ve”).

Envelope of Gram-positive



Envelope: Gram-negative bacteria



C) Capsule:

Polysaccharide structure surrounding the cell wall. Present in some G+ and G-ve.

Protects the bacteria from phagocytosis (hence, increase virulence), and can play a role in bacterial adherence to the host cell. Can serve as target for antibodies and may be used for making vaccines.

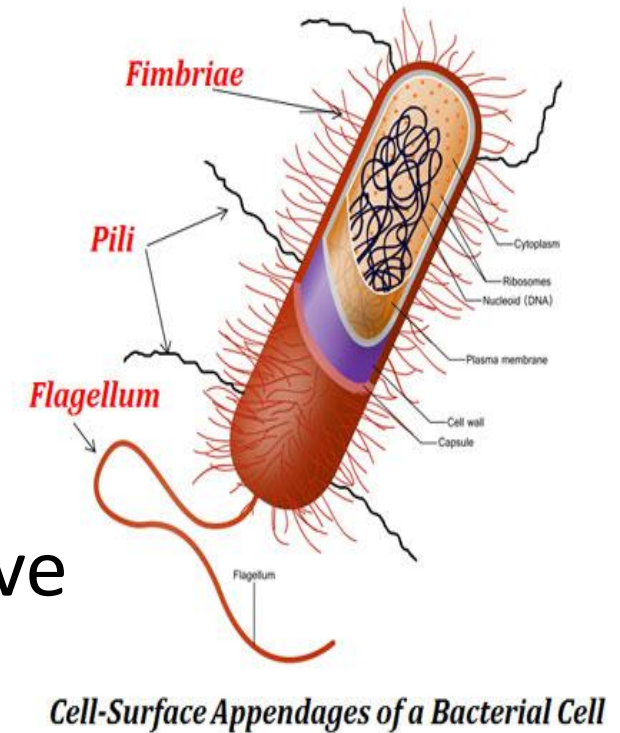
Appendages

Surface protrusions that include:

1. Flagella are protein appendages (flagellin protein). They are used locomotion and contain prominent antigenic determinants.

2. Pili or fimbriae, composed of pilin protein. They facilitate bacterial adherence to the host cells and serve as virulence factors in certain bacterial species.

3- Conjugation pili: rigid tubular structures used for DNA passage from one bacterial cell to the second.



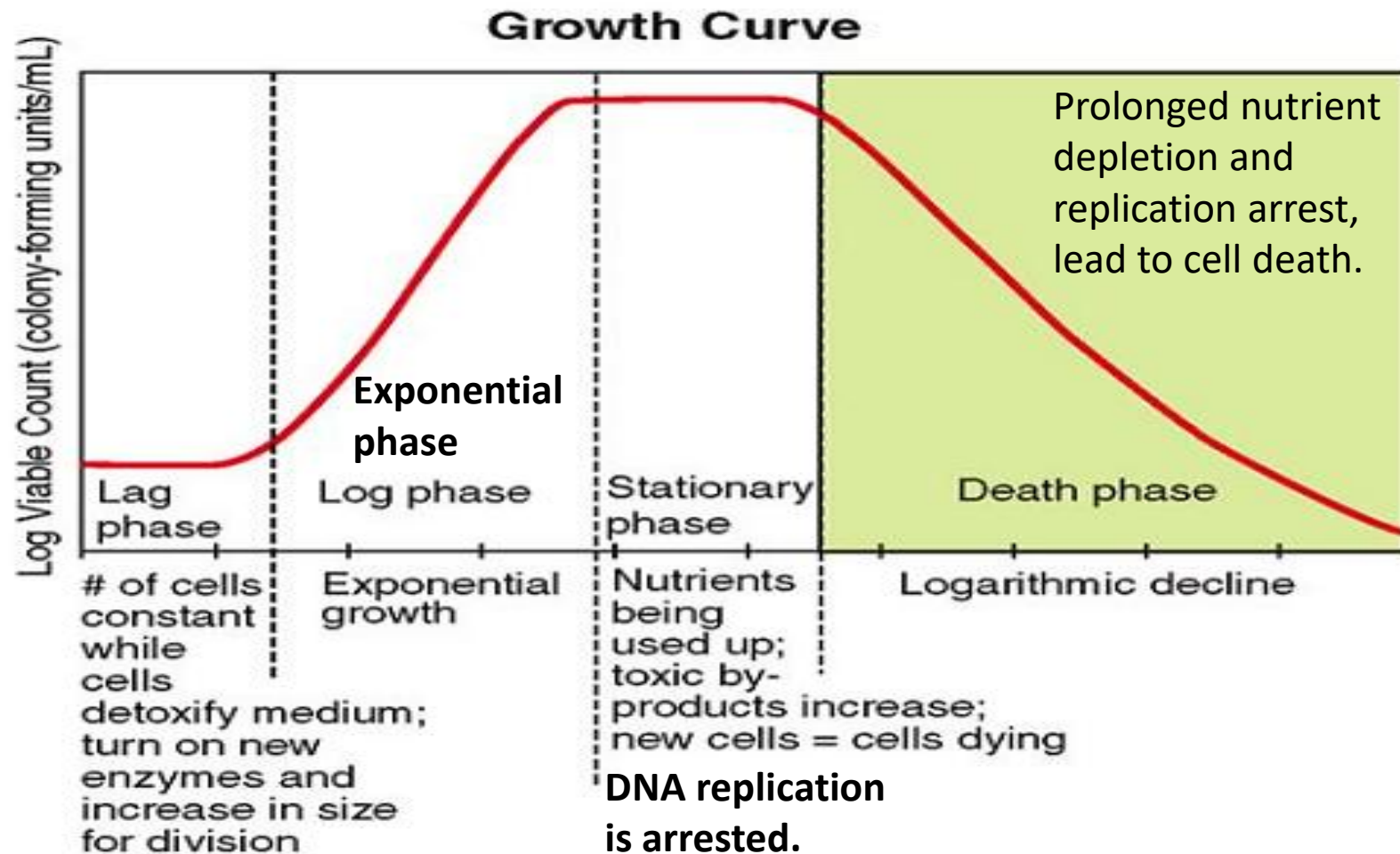
Endospores (spores)

- Survival response to protect the bacteria in extreme conditions.
- Composed of a core that contains DNA, ribosomes, large amount calcium dipicolinate, a cortex and a keratin-like coat.
- Endospores can survive for many years and germinate in favorable nutritional conditions. Calcium dipicolinate is essential for spore germination.
- Spore-forming bacteria include certain G +ve bacteria, like: *Bacillus* (aerobic) and *Clostridium* (anaerobic) species.

		CELL WALL TYPE ^a		
STRUCTURE	COMPOSITION	GRAM NEGATIVE	GRAM POSITIVE	NONE ^b
Envelope				
Capsule (slime layer)	Polysaccharide or polypeptide	+ or –	+ or –	–
Wall		+	+	–
Outer membrane	Proteins, phospholipids, and lipopolysaccharide	+	–	–
Peptidoglycan layer	Peptidoglycan (+ teichoic acid in gram positive)	+	+ ^c	–
Periplasm	Proteins and oligosaccharides in solution	+	–	–
Cell membrane	Proteins, phospholipids	+	+	+
Appendages				
Pili (fimbriae)	Protein (pilin)	+ or –	+ or –	–
Flagella	Proteins (flagellin plus others)	+ or –	+ or –	–
Core				
Cytosol	Polyribosomes, proteins, carbohydrates (glycogen)	+	+	+
Nucleoid	DNA with associated RNA and proteins	+	+	+
Plasmids	DNA	+ or –	+ or –	+ or –
Endospore				

Bacterial growth

Temperature, nutrient availability, pH and oxygen availability can affect growth curve. Significance: Antibiotics and dental plaque



Bacterial oxygen requirements

To generate ATP, bacteria can depend on respiratory (electron transport chain and oxygen) or fermentation (glucose & no oxygen) metabolism.

1- Strict/obligate aerobes: require oxygen for growth using respiratory metabolism, generating ROS (e.g: *M. tuberculosis*). They contain superoxide dismutase, which protects the bacterial cells from the ROS. Catalase that convert hydrogen peroxide to water and oxygen.

2- Strict/obligate anaerobes: grow efficiently in very low oxygen concentrations <1 %. Common in gut and mouth. Example: *Actinomyces*, *Clostridium* & *Fusobacterium*. Fermentation is the metabolic pathway (glucose to lactic acid, alcohol or different metabolites).

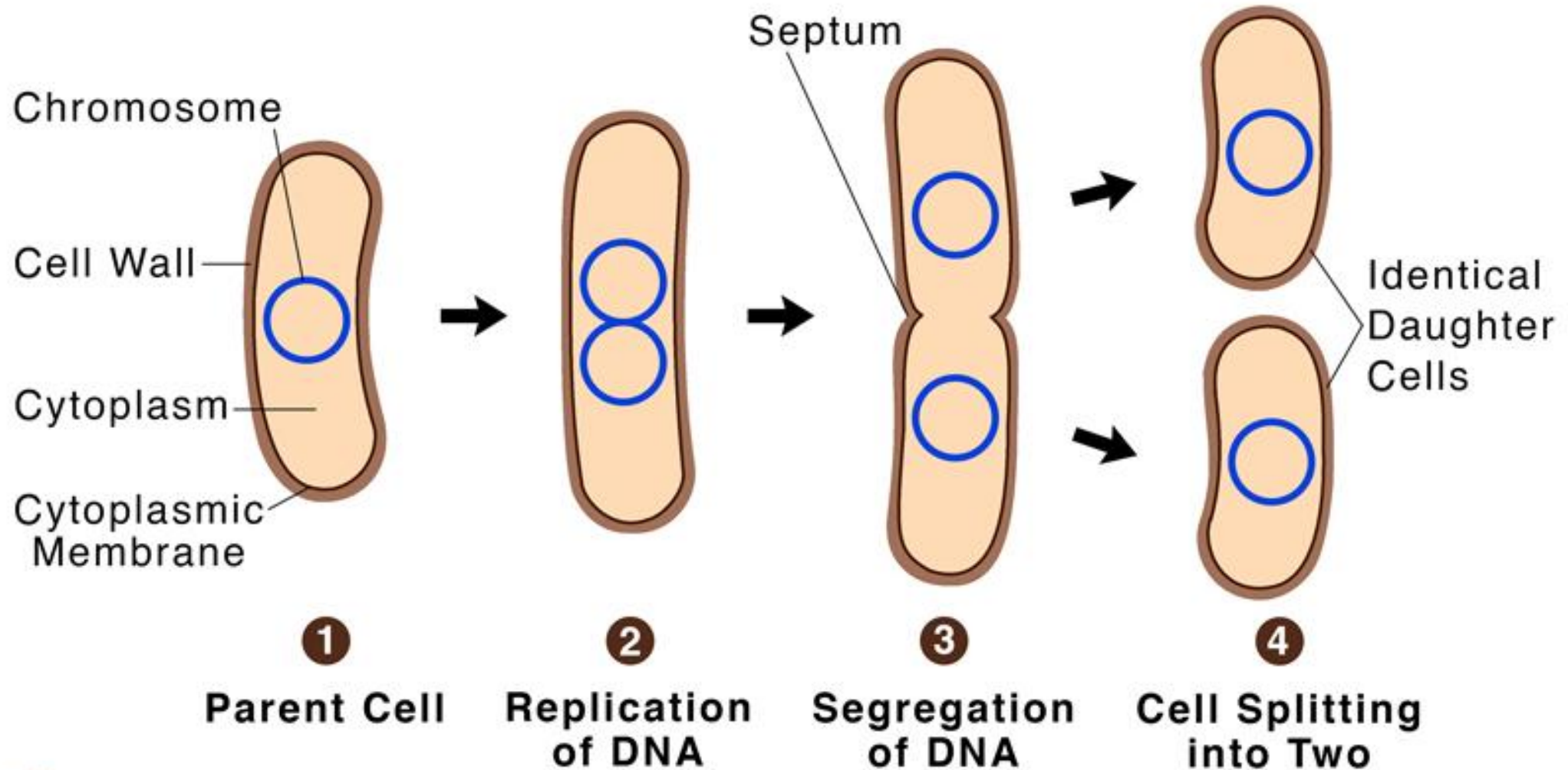
3- **Facultative anaerobes** (with/without oxygen). Most pathogenic bacteria are facultative anaerobes (*Staph.*, *Strep.*, *E. coli*). Shift from a fermentative to a respiratory metabolism in the presence of air.

4- **Aerotolerant anaerobes**: fermentative metabolism with and without oxygen.

5- **Microaerophiles**: require the presence of small amounts of oxygen. E.g, *Helicobacter sp.*

Bacterial replication

Binary fission



Bacteria can alternate between:

- **Planktonic**, free-floating individual bacterial cells.
- **Biofilm state**: immobile bacterial community. Aggregates of bacterial cells that colonize a surface are embedded in extracellular polymeric substance (EPS) matrix (polysaccharides, glycoproteins, lipids & DNA). Microbes usually live in communities.

Quorum sensing, occurs when individual species alter their gene expression to cooperate with other cells in the biofilm. Biofilm community enhance nutrient uptake and often decrease antibiotics susceptibility.

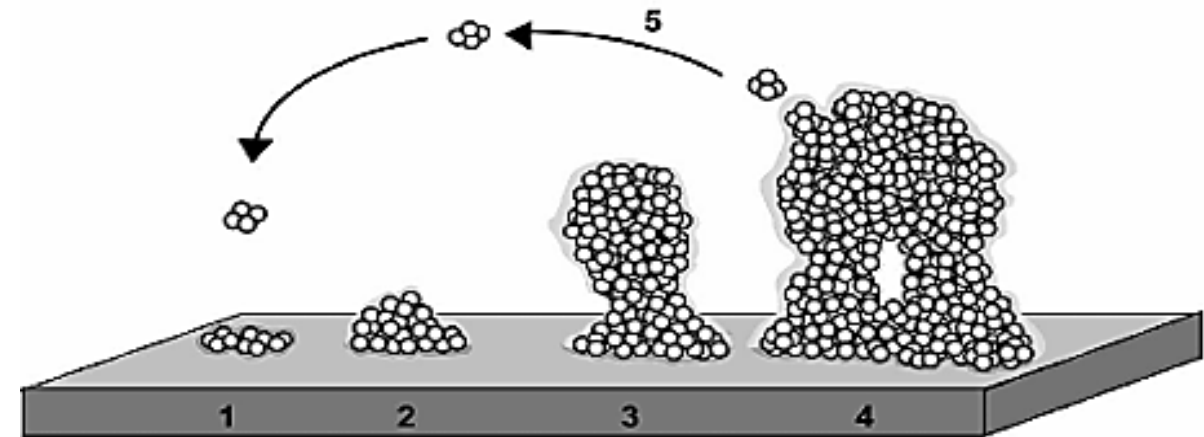
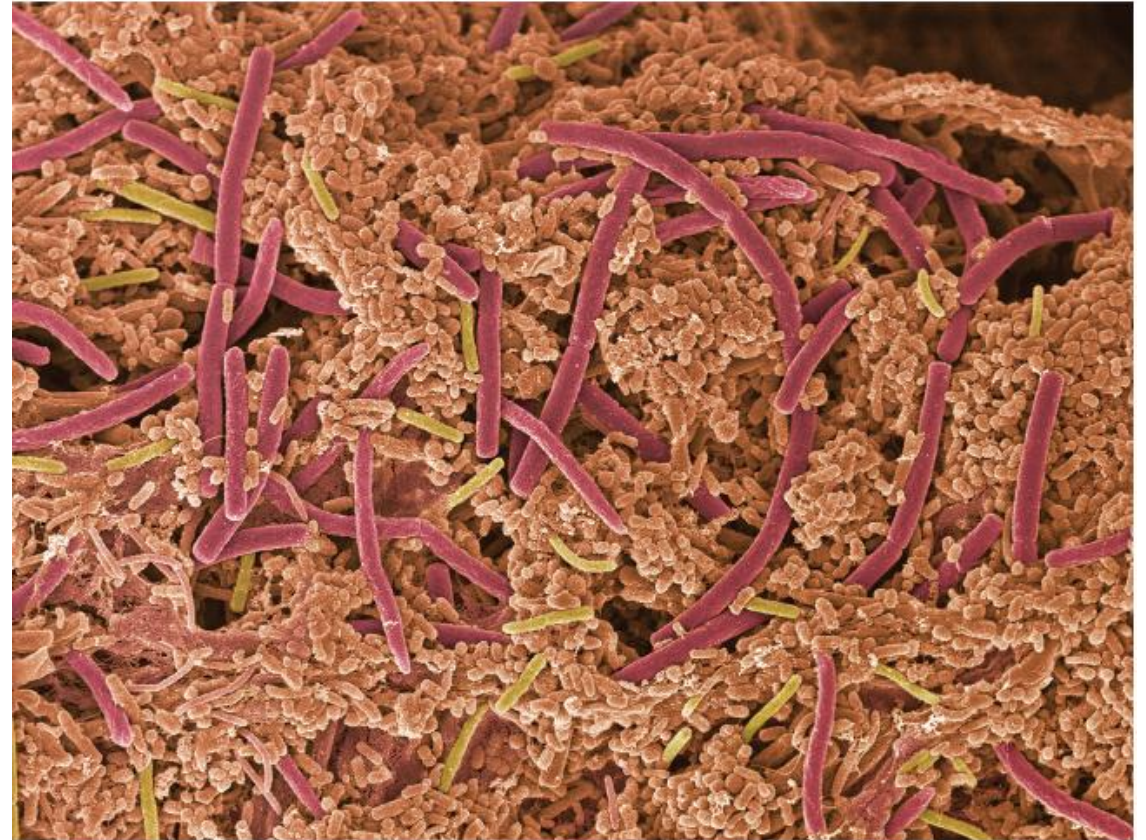


Figure 1: The development of a biofilm, depicted as a five-stage process.

Source: (Lasa, 2006) [30]

Stage1: *Reversible or Initial Attachment*: 2. *Irreversible Attachment*: 3. *Microcolony Formation*. 4. *Biofilm Maturation*. 5. *Biofilm Disper-*

Scanning electron
micrograph of a biofilm on
the surface of a tooth
(6,000×).



Steve Gschmeissner/Science Photo Library/Brand X
Pictures/Getty Images

Pathogenicity is, in a sense, a highly skilled trade, and only a tiny minority of all the numberless tons of microbes on the earth has ever involved itself in it; most bacteria are busy with their own business, browsing and recycling the rest of life. Indeed, pathogenicity often seems to me a sort of biological accident in which signals are misdirected by the microbe or misinterpreted by the host.

—Dr. Lewis Thomas, The Medusa and the Snail

Principles of Bacterial Pathogenesis

Pathogenicity: The capability of any bacterial species to cause disease in a susceptible human host.

- **Intercellular pathogens** should enter the cells and multiply in them e.g: *Mycobacterium tuberculosis* and *chlamydia*.
- **Extracellular pathogens:** don't invade cells. They proliferate in the extracellular environment e.g *Vibrio cholera*.

Virulence: to define the degree of pathogenicity. It is measured by the number of organisms required to cause disease. The infectious dose of an organism required to cause disease varies greatly among pathogenic bacteria.

Primary pathogens tend to cause disease in healthy hosts regardless of immune system status. Examples include *Helicobacter pylori*, *Mycobacterium tuberculosis*, *Salmonella sp.*

Opportunistic pathogens are pathogens capable of causing disease if the host has weakened immune system or altered microbiota. Normal microbiota can maintain the microbiome balance and prevent overgrowth. Examples include *Streptococcus sp.*, *Staphylococcus aureus*, *Clostridium sp.*, some strains of *E. coli*.

Normal flora: some nonpathogenic bacterial can live in/on humans for long periods without causing disease. **Dysbiosis** refers to microbial imbalance between the different microbial populations inside the body. It can contribute to pathological conditions.

Stages of pathogenicity

1- Exposure

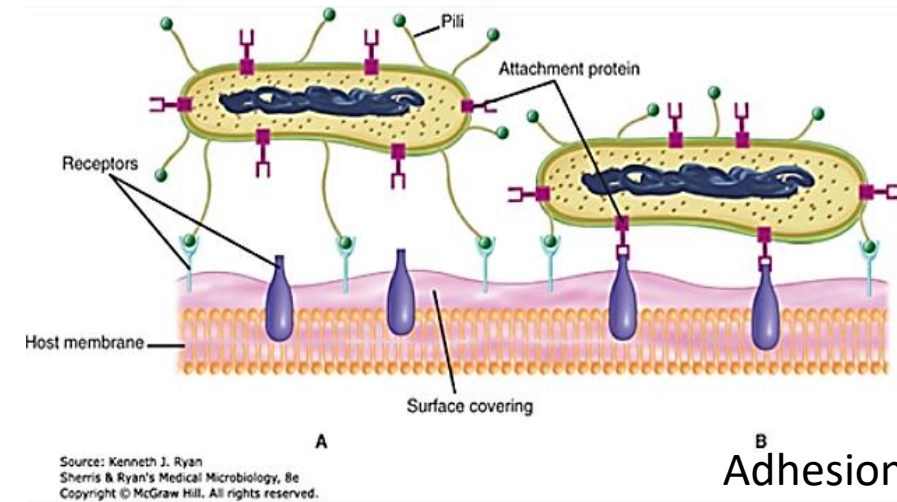
Pathogen should overcome the host barriers and gain access to the host tissue. Mucosal surfaces are important port of entry for microbes.

2- Adhesion

Several mechanisms. Pili or fimbriae or capsule facilitates bacterial adherence to the host cells.

Gram-negative bacteria membrane is rich with adhesins proteins.

Following adherence, bacteria can form biofilm for protection against antibiotics and host immune defenses.



3- Invasion

During invasion, pathogen aims to colonize the local tissue and evade the immune system. Several enzymes are secreted to aid in pathogenesis.

a- **Collagenase** Degrade the extracellular matrix by degrading collagen.

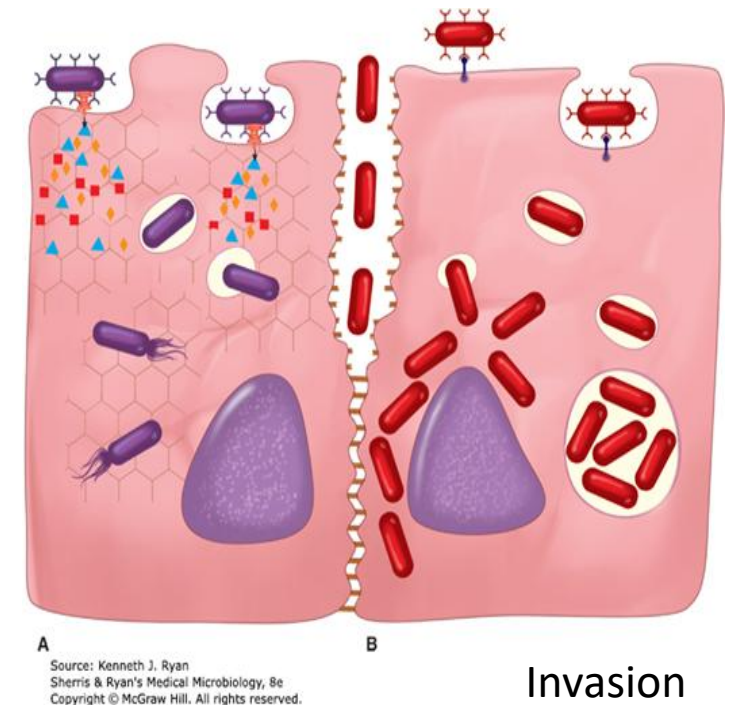
b- **Coagulase** converts fibrinogen to fibrin.

The insoluble fibrin fibers can protect the bacterial from phagocytosis.

c- **Immunoglobulin (Ig) proteases.**

IgA proteases inactivate IgA at the mucosal surface.

S. pneumonia can cleave IgG heavy chains to decrease the immune response.



4- Infection:

Successful multiplication of the pathogen leads to infection.

Local infection is confined to a small area.

Systemic infection refers to an infection that spreads in the whole body.

5- Disease transmission occurs when the pathogen moves from one host to the second. Mucus membranes are common ports of transmission.

Virulence factors

A) Structural bacterial products (antiphagocytic capsules, endotoxins and IgG-binding proteins).

B) Extracellular gene products (collagenase, IgA-degrading proteases, exotoxins, etc).

Collagenase is as a virulence factor for Clostridium histolyticum, facilitating the spread of gas gangrene. Medically, Collagenase from C. histolyticum. (brand name Santyl®) is used as a debriding agent in burn victims to help with healing.

c) Growth properties (intracellular growth and biofilms).

Bacterial toxins

Endotoxin

Are the lipid A component of the G –ve bacteria (LPS).

- a. Induce the release of cytokines.
- b. Increase vascular permeability.
- c. Activate the innate immune response.
- d. Cause fever, hypotension, disseminated intracellular coagulation, and shock.
- e. Causing ischemic or hemorrhagic necrosis in various tissues.

LPS

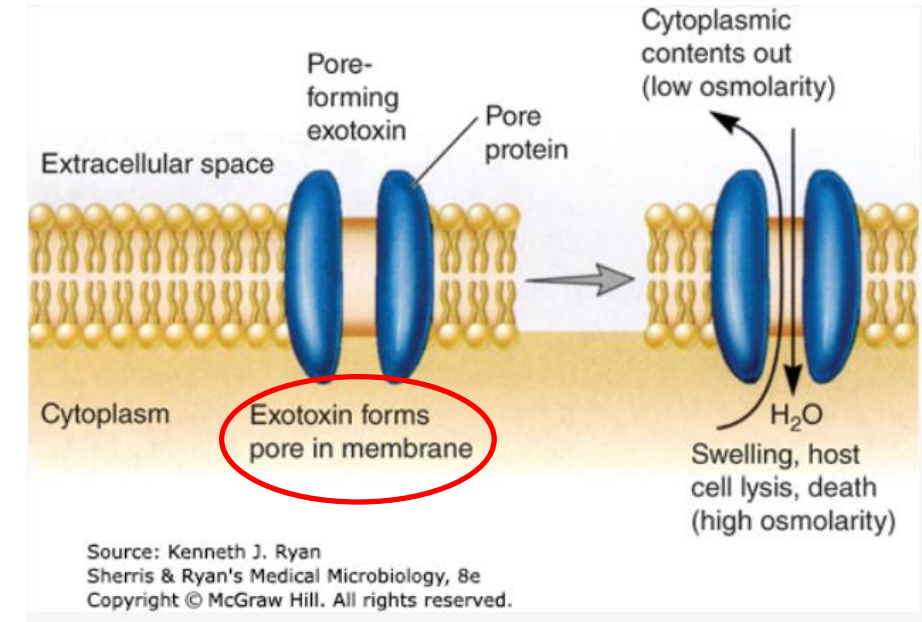


Exotoxins (G +ve & G -ve)

1- Pore-forming exotoxin

Disrupt the host cell membranes.

- *Staph. aureus* secretes alpha toxin that disrupts the cell membrane.
- *Streptococcus pyogenes* secrete streptolysin which create pores in the cell membrane.

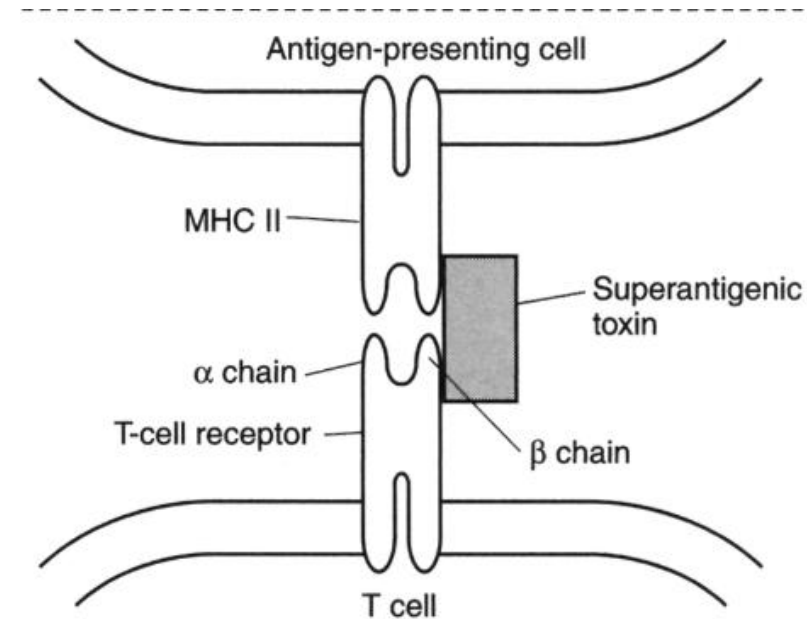


(2) Act as superantigens:

Cause massive release of cytokines.

S. aureus secrete toxic shock syndrome toxin (TSST-1) and enterotoxin superantigen that cause cytokine release. After the toxin is ingested, and absorbed in the gut, it causes vomiting and inflammatory cascade.

S. pyogenes secrete the erythrogenic toxins A and C. Those superantigens bind directly to MHC II, cause the recruitment of massive number of T cells and massive release of IL-2, TNF α and other inflammatory mediators leading to fever and shock.

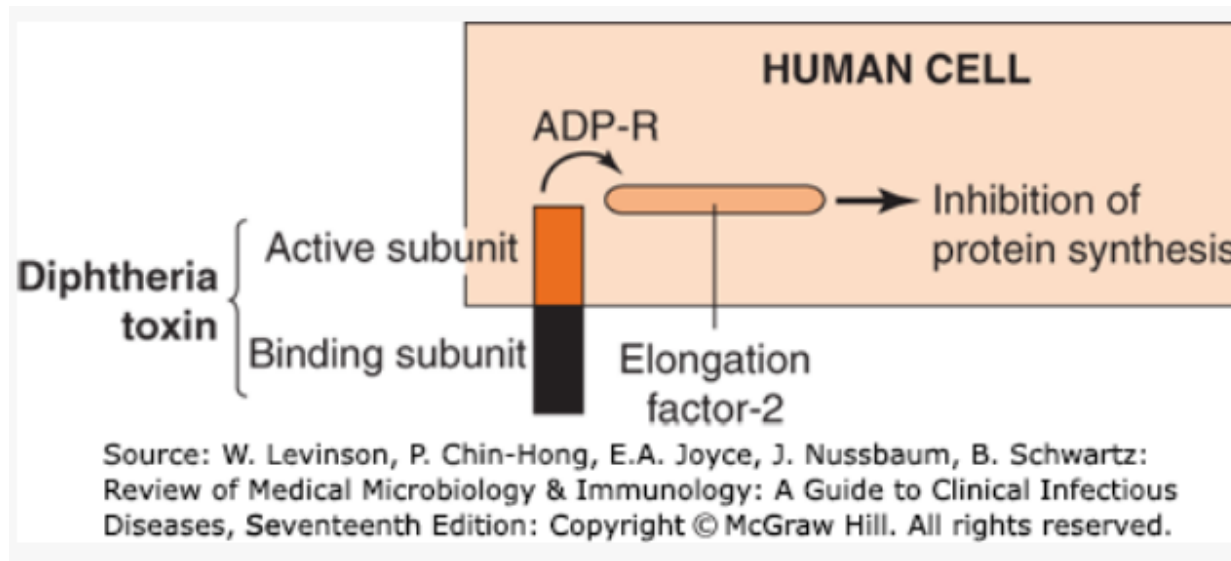


(3) Inhibit protein synthesis.

Corynebacterium diphtheriae secrete diphtheria toxin can inhibit the protein translation.

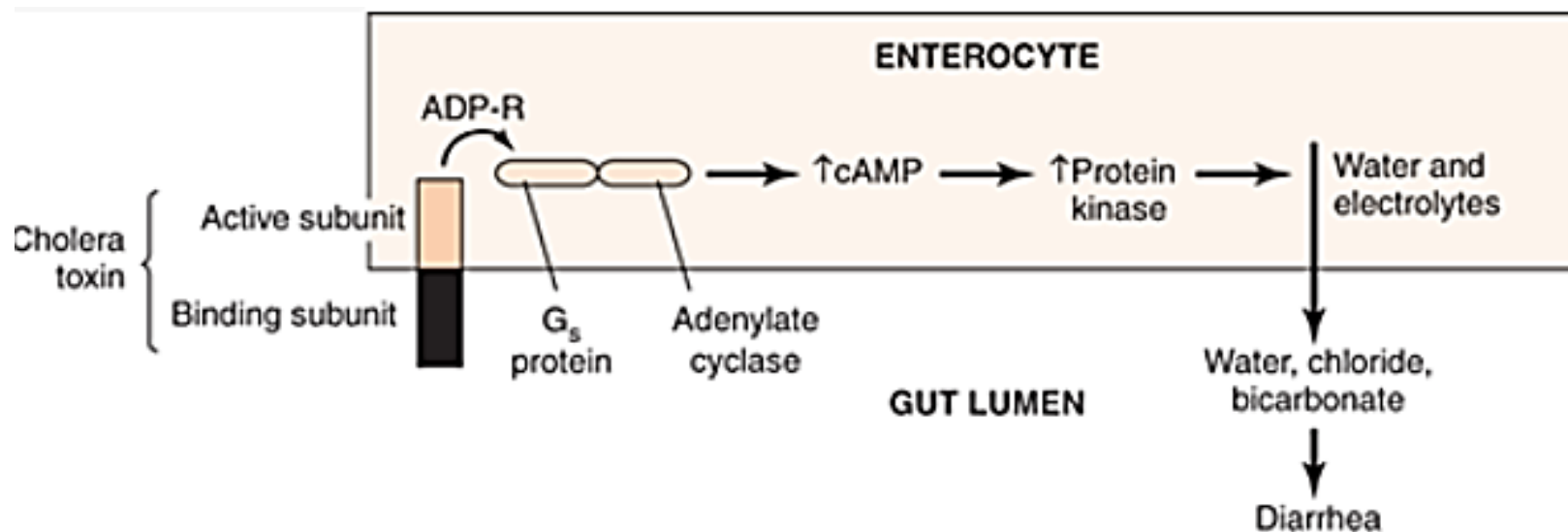
Shigella dysenteriae secretes Shiga toxin that inactivates the 60S ribosomes.

Enterohemorrhagic *E. coli* secretes vero toxin that also inactivate the 60S ribosomal subunit and inhibit the protein synthesis.



(4) Increased synthesis of cAMP

Cholera toxin from *Vibrio cholerae*. Cholera toxin upregulates protein kinase activity, increase electrolyte imbalance leading to watery diarrhea.

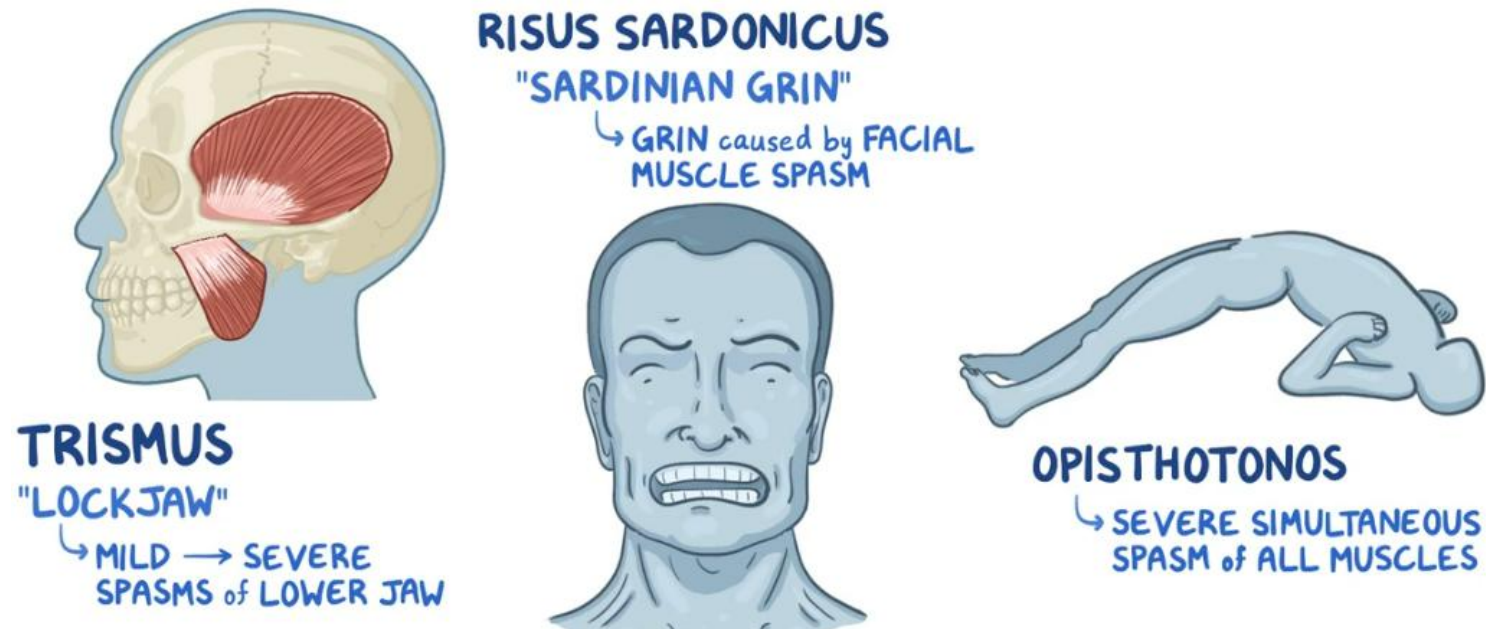


Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz:
Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious
Diseases, Seventeenth Edition: Copyright © McGraw Hill. All rights reserved.

(5) Alter nerve impulse transmission “neurotoxin”

- *Clostridium tetani* (Gram positive anerobic bacteria) secrete tetanus toxin (Tetanospasmin, rusty metals wounds). It induces spastic paralysis by interfering with the release GABA. Early symptoms include muscle rigidity, trismus, followed by sustained muscle rigidity and reflex spasm.

TETANIC TRIAD



- Botulism

Germination of *Clostridium botulinum* spores (Gram positive anerobic) in inappropriately canned food release botulinum toxin. Induce flaccid paralysis characterized by reduced muscle tone. Botulinum toxin inhibit the release of acetylcholine at cholinergic synapses in neuromuscular junctions and autonomic nervous system.

2 TYPES of BOTULINUM TOXIN: A & B

* LESS TOXIC

* LONG LASTING NEUROMUSCULAR JUNCTION BLOCK

↳ injected into rigid muscle to relax it

**OTHER DISEASES IMPROVED
with BOTULISM TOXIN:**

"BOTOX"

* HYPERHYDROSIS

* ACHALASIA

* FOCAL DYSTONIA



**Floppy baby syndrome
or infant botulism**



Exotoxins	Endotoxins
Secreted by the living cells	Part of the cell wall. Released after cell death.
Produced by Gram-positive and Gram-negative	Gram-negative only
Highly antigenic	Moderately or weakly antigenic
Can be converted to toxoids and used for immunization	Can not be converted to toxoids
Highly toxic	Moderately toxic
Fever less common	Usually produce fever.
Not heat stable. Can be denatured at 60 °C.	Heat stable, can withstand 60 °C heat for few hours.

Laboratory Diagnosis

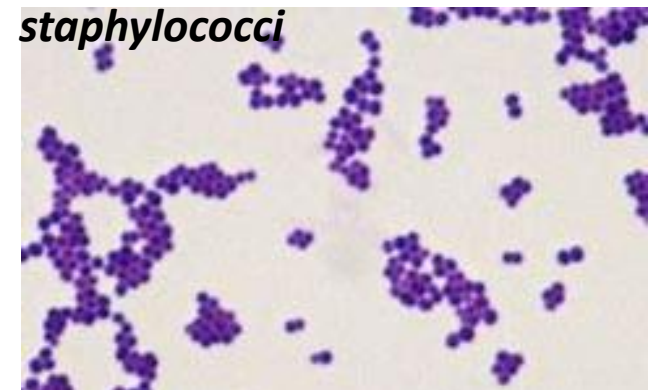
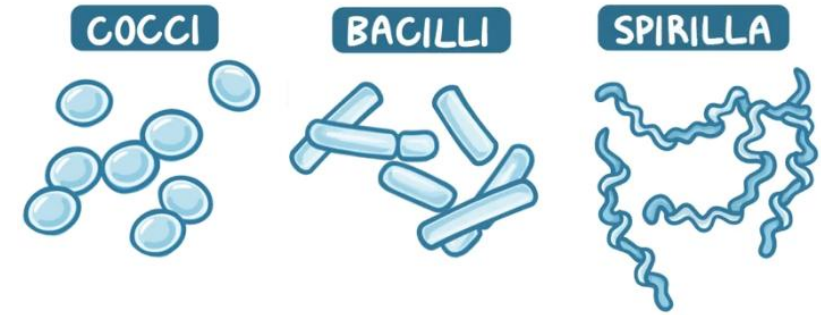
Bacterial Shape

a- Round “coccus” “diplo-, strepto-, staphylo-”

- Diplococci: often grow in doublets.
- Strepto: grow in chains “streptococci”
- Staphylo: remain in clusters “staphylococci”

b- Rod-like “bacillus”

c- Spiral “spirilla or spirochetes” (e.g: spirochetes are gram-negative, motile, spiral bacteria) like Treponema



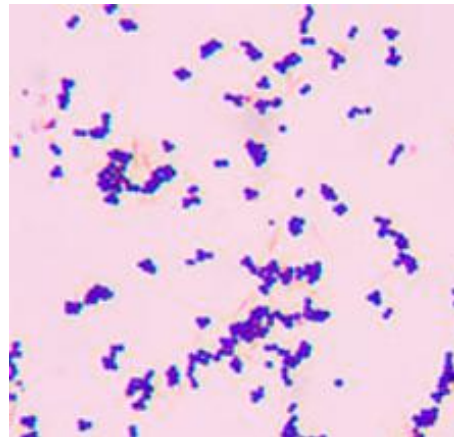
Staining

- Gram Stain (most cocci are G +ve, Bacilli are often G-ve)
- Giemsa for protozoa (malaria), intracellular microbes (chlamydia).
- Ziehl-Neelsen (acid-fast stain). e.g. *M. tuberculosis* bacilli. High lipid content of cell wall (mycolic acid) prevents Gram staining.
- Silver staining and crystal violet for *H. pylori*.

Gram staining

GRAM-POSITIVE

GRAM-NEGATIVE



Counter stain with
Safranin

Fixation



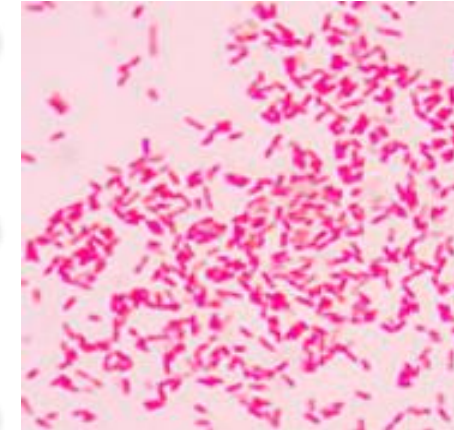
Crystal Violet



Iodine Treatment



Decolorisation



Come In and Stain!

r o l a
y d c f
s i o r
t n h a
a e o n
l l i n

V
i
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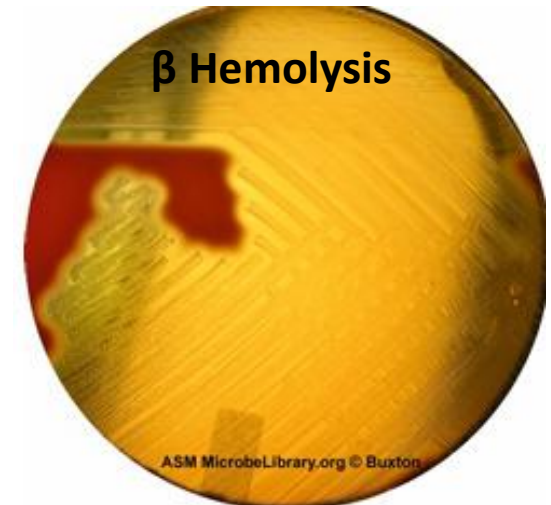
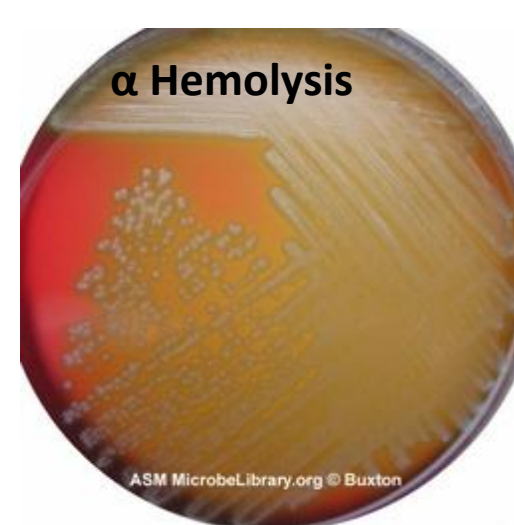
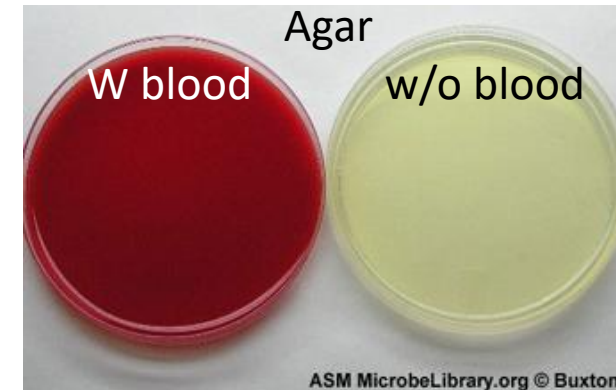
Gram-positive bacteria stains purple due to thick layer of peptidoglycan. While Gram-negative bacteria stains pink due to the thin layer of peptidoglycan and high lipid content.

Others

1- G +ve test:

a. Hemolytic pattern on blood agar.

Type	Reaction on Blood agar
No hemolysis (Gama-hemolysis)	No color change (reddish agar with bacterial colonies)
Alpha hemolysis	Partial lysis (greenish-brownish discoloration)
Beta-hemolysis	Complete lysis (Yellowish discoloration)



- Enzymes:
 - Catalase: enzyme that neutralizes the bactericidal effects of hydrogen peroxide. Defense mechanism to allow them escape its oxidative damage. Converts hydrogen peroxide into water and oxygen
 - Coagulase: For ***Staph. aureus*** (positive). Bacteria accelerates clot formation (mix bacterial cells with plasma). ***Staph.*** secrete staphylocoagulase that can initiate blood coagulation by activating prothrombin to form fibrin.
- Antibiotic sensitivities

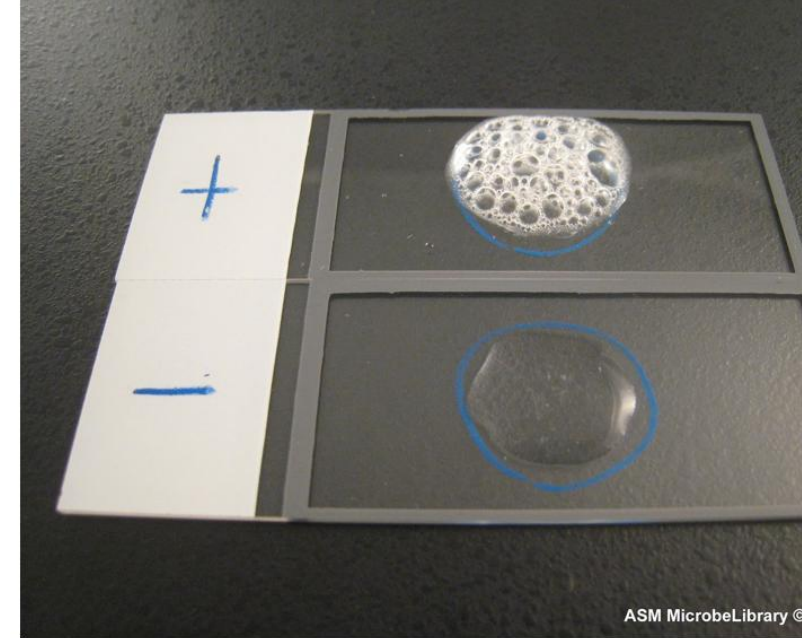
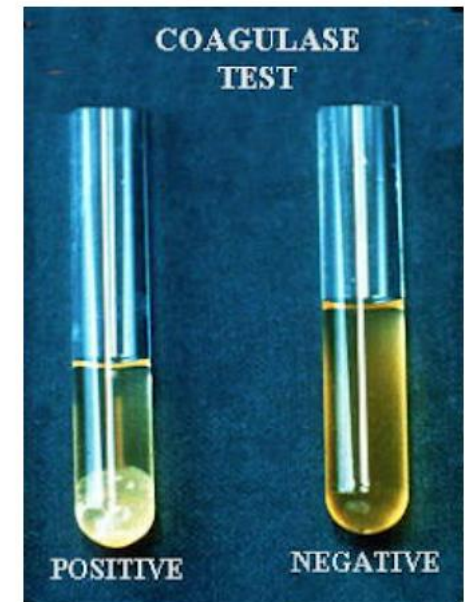


FIG. 1. Slide catalase test results. (Top) The positive reaction was produced by *Staphylococcus aureus*; (bottom) the negative reaction was produced by *Streptococcus pyogenes*.



Coagulase Test

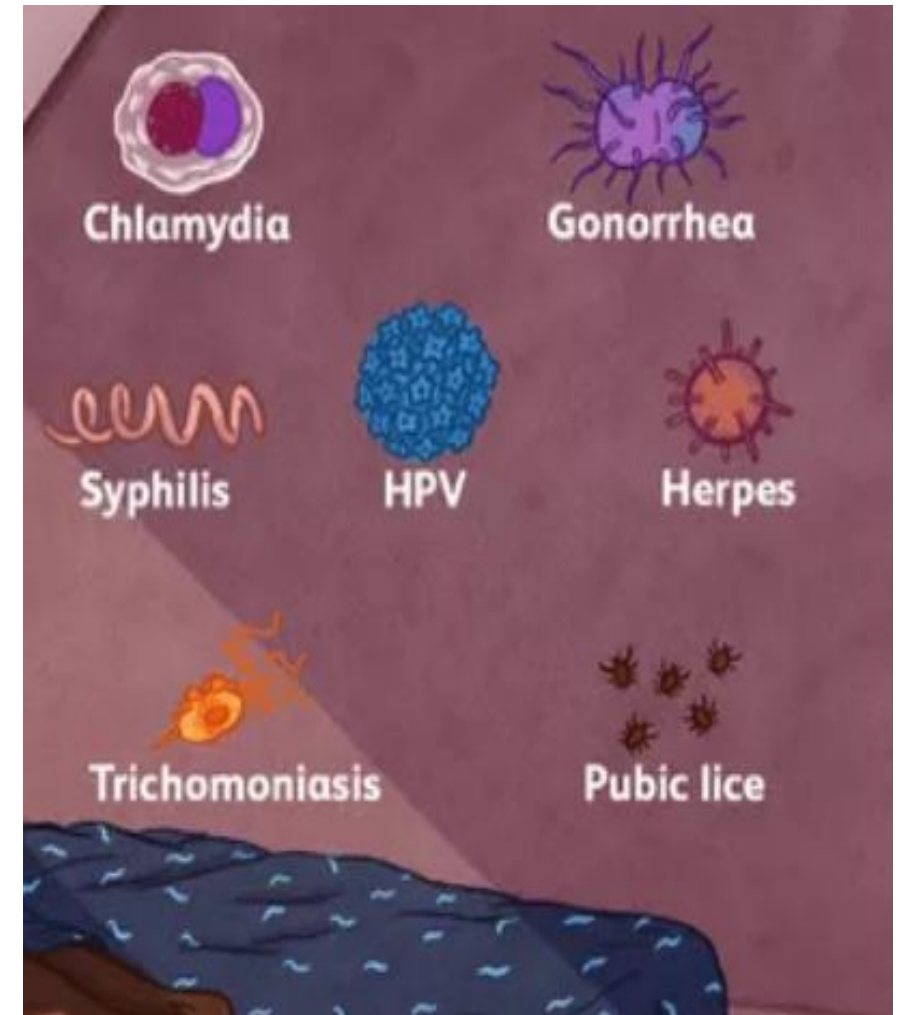
2- G -ve tests

- a. Enzymes like maltase, oxidase, urease
- b. Bacterial antigens:
 - i. K antigen is for the bacterial capsule. H antigen is on the flagellar protein (or bacterial flagella).
 - ii. The cell wall O antigen is the outer polysaccharide portion of LPS. The O antigen is highly variable. For example, more than 1500 different O antigens were identified in *Salmonella* genus and around 150 types were identified in *E coli*.

Bacteria involved in human disease

Sexually transmitted infections

- **Bacteria:** including chlamydia (*Chlamydia trachomatis*), gonorrhea (*Neisseria gonorrhoeae*), and syphilis (*Treponema pallidum*)
- **Viruses:** including HIV, herpes simplex virus, human papillomavirus, hepatitis B virus, cytomegalovirus (CMV), and Zika.
- **Parasites** like *trichomonas vaginalis*.
- **Insects** such as crab lice.



- ***Chlamydia trachomatis***

Chlamydia are obligate intracellular bacteria that can be stained with Giemsa. They are the most common (bacterial) cause of sexually transmitted infections in US. It causes genital tract infections. Reside in the human genital tract and eyes. Transmission is by sexual contact and during vaginal delivery. There are 15 serotypes of *C. trachomatis*. Certain serotypes can cause trachoma (eye infection or chlamydia conjungivitis, transmitted by hand-to-eye contact) or pneumonia (respiratory infection).

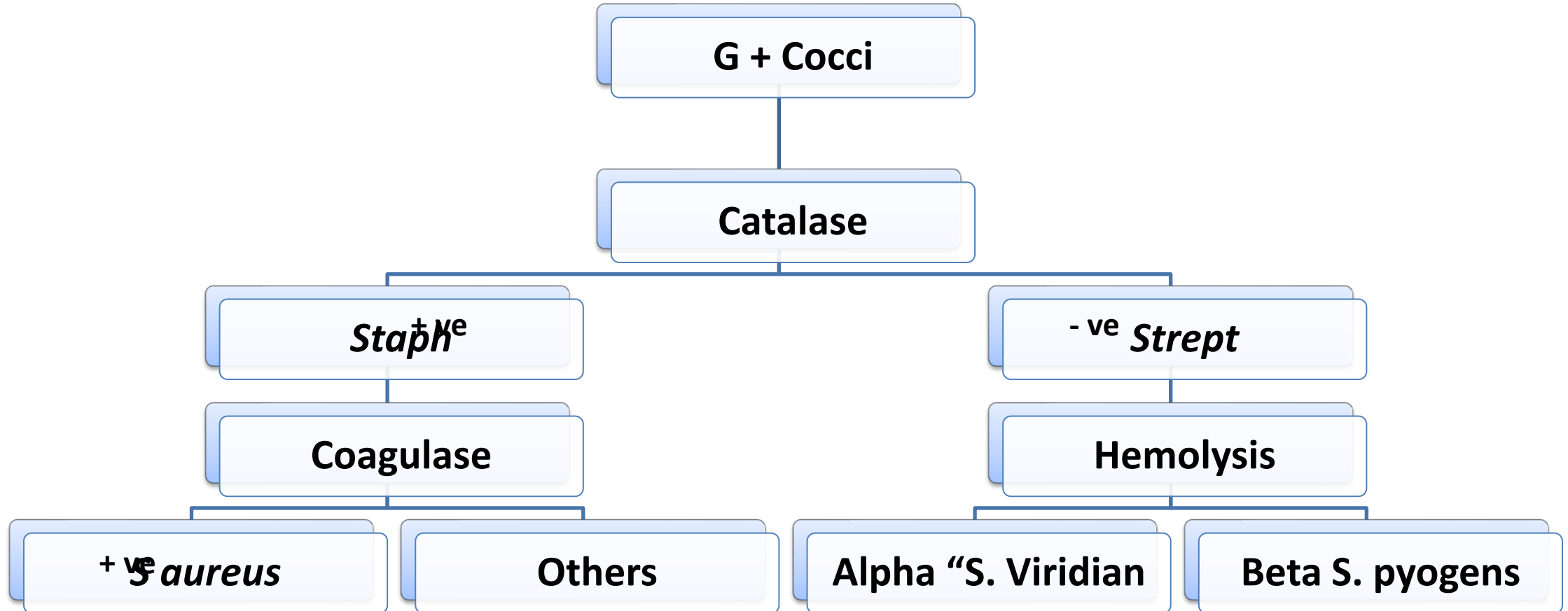
- ***Neisseria gonorrhoeae* (Gonococcus)**

This is Gram-negative diplococci that cause gonorrhea (symptoms vary between male and females but they share inflammation, discharge and pain during urination) and neonatal conjunctivitis and pelvic inflammatory disease. The bacteria invades mucous membranes and causes inflammation. They reside in the human genital tract and eyes. Transmission: sexual contact and during vaginal delivery.

- ***Treponema pallidum***

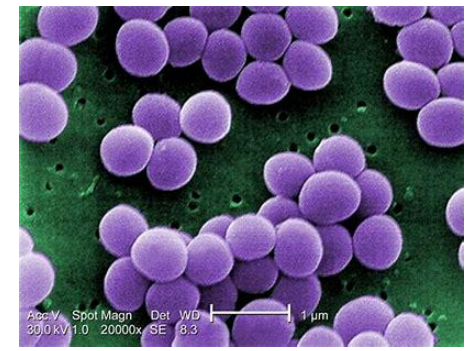
These are spirochete shaped bacteria. It causes syphilis that can progress from enlarged lymph nodes near the groin to organ damage and even death. They are not Gram-stained because the bacteria is too thin. Many features of syphilis are attributed to blood vessel involvement causing vasculitis. Reside in the human genital tract and eyes. Transmission: sexual contact, vaginal delivery and across the placenta.

Gram Positive Bacteria



Catalase test differentiates staphylococci (catalase positive) from streptococci (negative).

Staphylococcus aureus



- The genus *Staphylococcus* has at least 45 species. *S. aureus* is an important cause of abscess.
- **Features:** Gram-positive facultative intracellular cocci that grow in clusters “grape-like.” *S. aureus* is coagulase-positive and catalase-positive with antiphagocytic capsule, β-hemolytic, yellow colonies on blood agar. Most isolates produce β-lactamase conferring antibiotic resistance.
- **Habitat/transmission:** Nose and skin. Transmission via direct contact (hands) or inhalation.

1- *Coagulase*: *S. aureus* produces an extracellular coagulase which clots the citrated plasma.

2- *Beta lactamase enzyme* is expressed by the plasmid genes. Causes penicillin antibiotic resistance

S. aureus

Virulence factors:

1- Toxic shock syndrome toxin (TSST) is a superantigen and can causes toxic shock syndrome by stimulating cytokine release **from helper T cells, especially interleukin-2.**

2- Enterotoxin, another a superantigen that causes food poisoning,

3- Protein A binds to the heavy chain of IgG, leading to decreased phagocytosis

4- Beta lactamase causes penicillin antibiotic resistance.

5- Exfoliative toxin A: destroys the attachment of keratinocytes in the epidermis (leading to sloughing and honey-colored blistering skin lesions).

MAJOR VIRULENCE FACTORS	DISEASE
Protein A	Abscess
β -lactamase	Pneumonia
Enterotoxin	Toxic shock
TSST	syndrome
Exfoliatin	Scalded skin
Hyaluronidase	syndrome
Staphylokinase	Food poisoning
	MRSA
	Endocarditis
	Osteomyelitis



Diseases: common cause of human infections.

- Abscesses of skin and soft tissue infection.
- Impetigo (bacterial skin infection, usually Staph or Step)
- Acute endocarditis (rapid onset).
- Can cause rapid-onset food poisoning.
- Staph can also create biofilm on medical implants like prosthetic heart valve and artificial joints.
- Cause hospital-acquired pneumonia, surgical wound infections, and sepsis.
- Cellulitis



Methicillin-resistant *Staphylococcus aureus* (MRSA) is the most common cause of skin abscesses in the United States. Two subtypes HA-MRSA (health-care associated) and CA-MRSA (community associated). They are resistant to all β -lactam antibiotics (respond to vancomycin). There are vancomycin-resistant *S. aureus* (VRSA, rare so far).

Staphylococcus epidermidis is a Gram-positive, facultative anaerobic, catalase positive, coagulase negative and gamma hemolytic. Dominant normal skin flora. More significant role in contaminating and making biofilms on permanent medical devices like prosthetic joints.

Streptococci sp.

Lancefield grouping

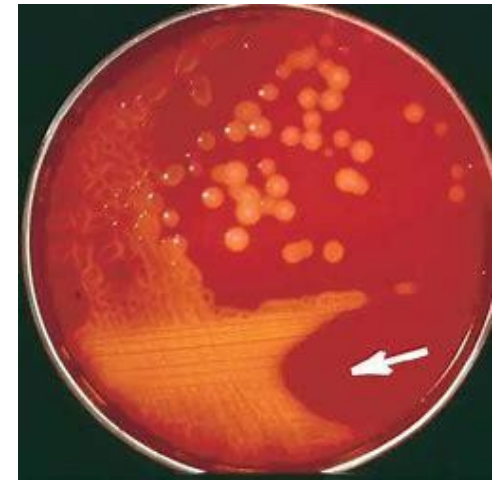
- *Streptococci* sp. is classified according to the hemolytic properties into alpha and beta.
- β -hemolytic streptococci are further classified by the Lancefield grouping into 21 classes designated by letters (A to H & K to U) based on the presence of polysaccharide and teichoic acid antigens in the bacterial cell wall .
- Medically important streptococci belong to group A and B.
- *Strept.* are catalase –ve.

1- *Streptococcus pyogenes*

Gram-positive cocci, β -Hemolytic, facultative anaerobic, encapsulated bacteria (Group A). Resides in the human throat and skin, transmitted via respiratory droplets. Human-specific bacterial pathogen that can cause mild localized infections to life-threatening invasive infections.

Virulence factors:

- M protein is a virulence factor (inhibit phagocytosis).
- Exotoxin A (superantigen)
- Hyaluronidase and collagenase (facilitates spread in tissue).
- Hemolytic streptolysin S and O exotoxin (cytotoxic).



Diseases:

1. Pyogenic (aka, suppurative or pus-producing) diseases like gp A strep. pharyngitis (Strep throat) in the oropharynx (sore throat and fever, more common in children <15 Y), cellulitis, impetigo. “M protein”
2. Toxigenic diseases like necrotizing fasciitis or “flesh-eating” streptococci and scarlet fever and Streptococcal toxic shock syndrome (STSS).
3. Immune-mediated like rheumatic fever. “M protein”



Streptococcal toxic shock syndrome (STSS), acute-onset severe illness caused by *S. pyogenes*. *Symptoms include fever, myalgia, nausea*, can progress rapidly to sepsis with hypotension, tachycardia and organ failure, which may lead to mortality (**30 %-50 %**). Pathophysiology: production of bacterial superantigen exotoxins lead to the over-activation of cytokines and inflammatory cells.

2- *Streptococcus agalactiae*

These are Gram-positive cocci, encapsulated, β -Hemolytic bacteria (group B). They reside in the human vagina. They cause neonatal meningitis and sepsis.

3- *Streptococcus pneumoniae*.

Lancet-shaped, gram-positive, alpha hemolytic, encapsulated bacteria. Most *S. pneumoniae* serotypes can cause disease like pneumonia and meningitis in adults and otitis media and sinusitis in children. Produce IgA protease.

Viridan *Streptococci*

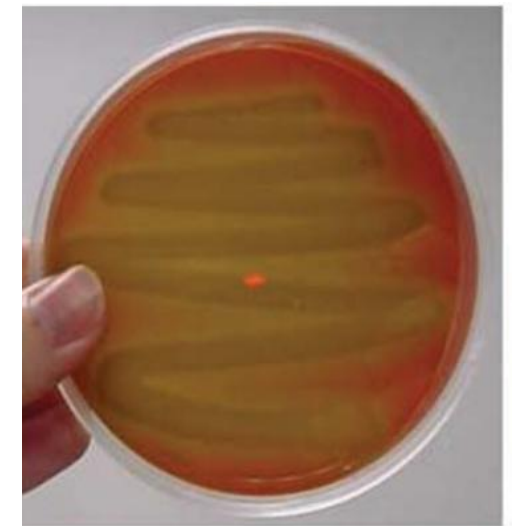
(*S. mutans*, *S. mitis*, *S. anginosus*, *S. sanguis* and *S. salivarius*)

Characteristics: Gram-positive cocci growing in chains. α -hemolytic (greenish colonies on blood agar). Catalase-negative.

Habitat: normal flora of oropharynx.

Diseases: *S. mutans* and *mitis* adhere to the teeth forming the dental plaque, implicated in dental caries.

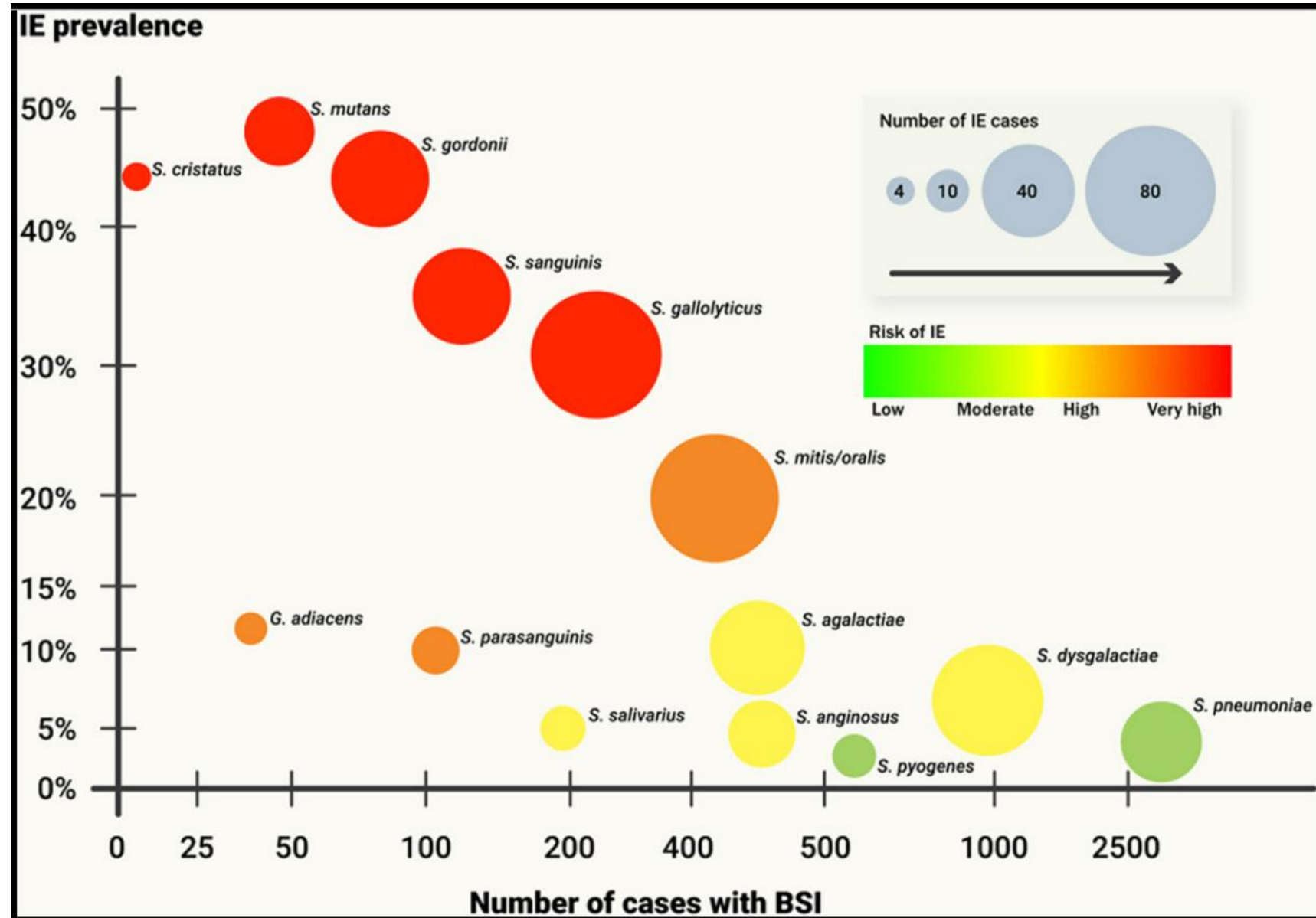
Viridan streptococci can enter the bloodstream and cause bacteremia, which may cause subacute endocarditis (*S. sanguis*, mitral valve, slower onset).



Alpha Hemolysis

- **Pathogenesis:** low virulence organism with no known toxins. Glycocalyx (glycoprotein in the cell membrane) is composed of polysaccharide that enhances adhesion to heart valves and tooth enamel.
- **Treatment:** Penicillin group. Patients with prosthetic heart valves take antibiotic prophylaxis to avoid infective endocarditis before invasive dental procedures (2 grams administered 30-60 min before the procedure).

Prevalence of infective endocarditis in streptococcal blood stream infections is species dependent. Highest associated IE risk observed with *S mutans*, *S gordonii*, *S sanguinis*, and *S mitis/oralis*.



Enterococci

Habitat :

Normal gut microbiota; Also, isolated from endodontic infections. Difficult to eradicate due to persistence within dentinal tubules. *E faecalis* can be associated with treatment failure in endodontic infections.

Characteristics:

G +ve facultative anerobic bacteria, non-spore forming.

Enterococcus faecalis is the most virulent species.

Can cause subacute infective endocarditis in hospitalized patients , urinary tract infections.

Anerobic infection: *Clostridium difficile*

Characteristics:

Obligate anaerobic Gram-positive spore-forming bacilli.

Habitat and Transmission:

Environment, reside in the human colon following ingestion of spores.

Fecal-oral route.

Over-proliferate upon disruption of normal flora upon use of broad-spectrum antibiotics or chemotherapy.

Disease: cause pseudomembranous colitis via its enterotoxin toxin A and the Cytotoxin, toxin B leading to diarrhea, abdominal pain and fever. Untreated condition may progress to colon rupture and sepsis.

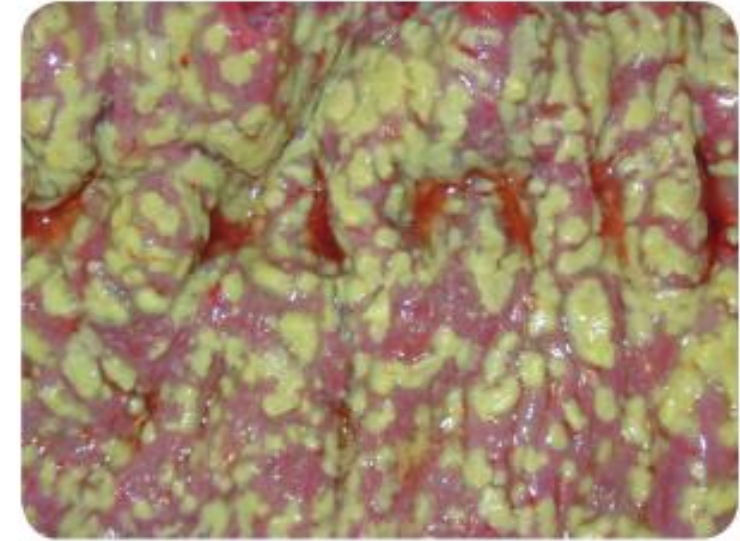


Figure 56.1 The gross pathological appearance of the colonic mucosa in pseudomembranous colitis.

Yellowish plaque “pseudomembrane” inflammatory response and pus

Treatment:

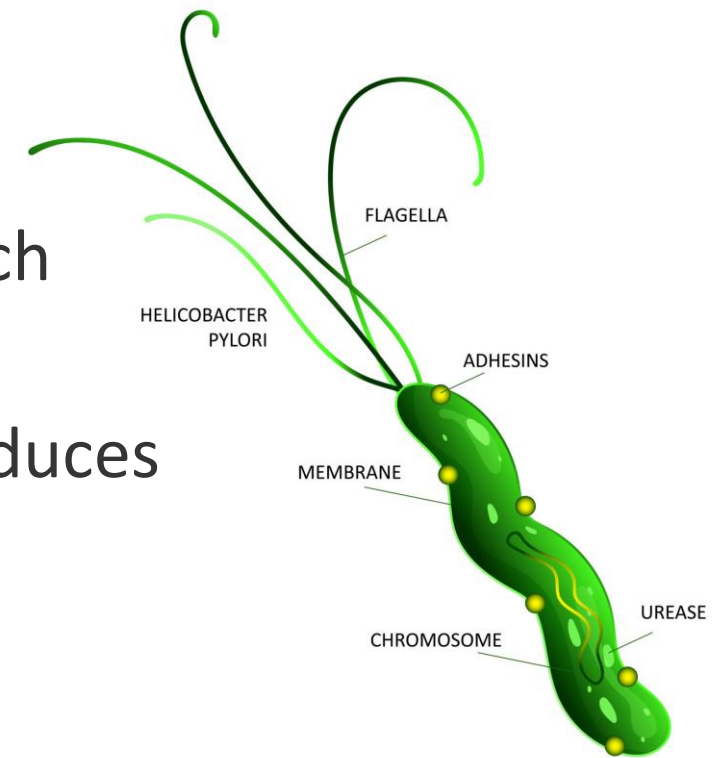
Antibiotics metronidazole (mild to moderate cases) and vancomycin (severe disease).

Fecal transplant along with probiotics may be an alternative treatment for recurrent infections.

2. Gram Negative Bacteria

Helicobacter pylori

- **Diseases:** Gastritis and peptic ulcer. Risk factor for gastric carcinoma.
- **Characteristics:** Curved gram-negative bacilli.
- **Habitat and Transmission:** Habitat is the human stomach and transmission is by ingestion.
- **Pathogenesis:** Organisms synthesize urease, which produces ammonia that can damage gastric mucosa and protect bacteria from acidic environment.
- **Treatment:** Quadruple drug therapy (amoxicillin, clarithromycin, metronidazole and proton pump inhibitor).



Pseudomonas aeruginosa

- **Diseases:** multiple, pneumonia (cystic fibrosis patients), otitis externa, contaminate surgical wound or burn victims; bacteremia. Higher risk in older patients or immunocompromised patients.
- **Characteristics:** catalase positive, motile obligate aerobic bacilli.
- **Habitat:** Water resources and environment. Long survival. Transmitted via aerosols or non intact skin/ mucosal membranes exposure.
- **Pathogenesis:** exotoxin A and endotoxin “LPS” that cause fever and shock, phospholipase C that degrades the cell membranes.
- **Sig.:** often multi-antibiotic resistant and capable of forming biofilms. Treatment include anti-pseudomonal penicillin w lactamase inhibitor (piperacilline with tazobactam).

Enterobacteriaceae

```
graph TD; A[Enterobacteriaceae] --> B[Escherichia]; A --> C[Salmonella]; A --> D[Shigella]; A --> E[Klebsiella]; A --> F[Yersinia and others]; B --- G[Lactase fermenter]; C --- H[Non Lactase fermenter];
```

Escherichia

Lactase fermenter

Salmonella

Non Lactase fermenter

Shigella

Klebsiella

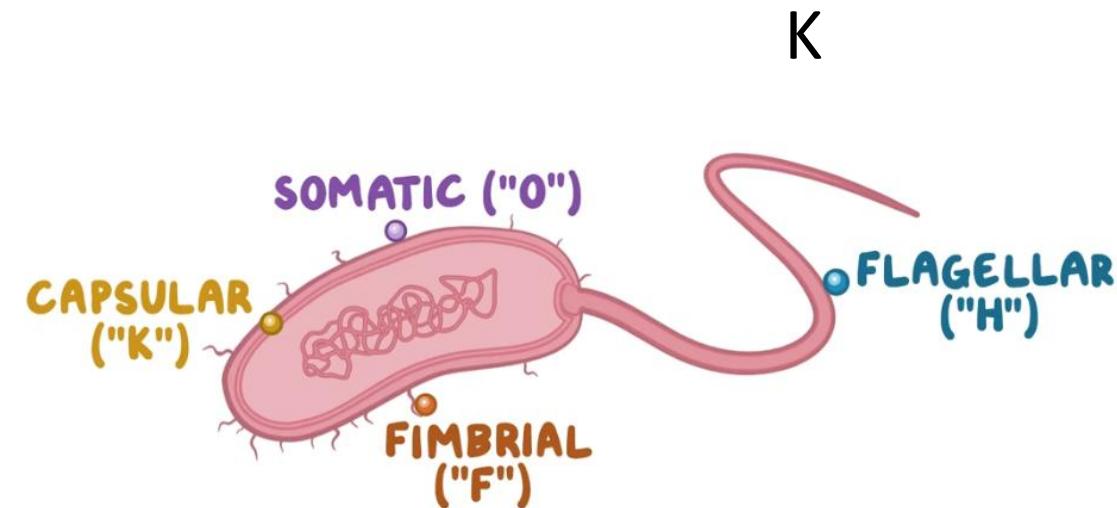
*Yersinia
and others*

Escherichia coli

Characteristics: Facultative Gram-negative bacilli with capsule, catalase positive.

Habitat and Transmission: Reside in the human colon; it colonizes the vagina and urethra (can cause urinary tract infection). Acquired during vaginal birth or by the fecal–oral route.

Antigenicity: O antigen (cell membrane),
(capsule), H (flagella), F(fimbria).
Used for classification of different subtypes.



1- Shiga-like Toxin Producing E.coli (STEC)

O157:H7 strains of *E. coli*. Causes bloody diarrhea and hemolytic-uremic syndrome (HUS). Toxin damage the glomeruli causing proteinuria.

2- Enterotoxigenic *E. coli* (ETEC) strain is associated with traveler's diarrhea "watery, non-bloody diarrhea. It is self-limiting, and of short duration (1–3 days).

3- Enteroinvasive E coli, or EIEC, causes bloody diarrhea by mechanically damaging the intestinal epithelium.

4- Enteropathogenic *E coli*, or EPEC.

Causes watery diarrhea in children <2 years old.

5- Uropathogenic *E coli*, or UPEC.

Mainly responsible for the development of urinary tract infections in females.

SYMPTOMS

STEC

* DIARRHEA



ABDOMINAL CRAMPS
& VOMITING

* BLOODY DIARRHEA

* ↓-GRADE FEVER

HEMOLYTIC-UREMIC
SYNDROME

* body swelling

* confusion

* nausea

* vomiting

* PALE

CONJUNCTIVA

* JAUNDICE

* DIARRHEA

* ABDOMINAL PAIN

ETEC

* DIARRHEA



ABDOMINAL CRAMPS
& VOMITING

* NON-BLOODY
DIARRHEA

* WATERY DIARRHEA

* FEVER

* BLOATING

EIEC

* DIARRHEA



ABDOMINAL CRAMPS
& VOMITING

* BLOODY DIARRHEA

* CHILLS

EPEC

* DIARRHEA



ABDOMINAL CRAMPS
& VOMITING

* NON-BLOODY
DIARRHEA

* SEVERE
DEHYDRATION in
CHILDREN
UNDER 2

* PROLONGED or
CHRONIC DIARRHEA



MALNUTRITION or
STUNTED GROWTH

UPEC

UTI SYMPTOMS:

* DYSURIA

* URINARY
FREQUENCY

* FLANK PAIN

Salmonella enteritidis

Diseases: Enterocolitis.

Characteristics: Gram-negative facultative bacilli (non-capsulated)

Habitat and Transmission: GI tract. Transmission by the fecal–oral route.

Pathogenesis: Has endotoxin & can secrete exotoxins (cytotoxin and enterotoxin). Invades the mucosa of the small and large intestines. Can enter blood, causing sepsis.

Symptoms: Start after 24-48 h, watery diarrhea, abdominal cramps, nausea, vomiting, low-grade fever, resolve in 2 days.

Salmonella typhi

Diseases: Typhoid fever endemic in certain areas (Typhoid vaccine).

Characteristics: Encapsulated Gram-negative facultative bacilli.

Habitat and Transmission: GI tract, transmission fecal-oral route. Bacteria reside in gall bladder & is released to intestine and feces.

Pathogenesis: expresses virulence (Vi) antigen on its polysaccharide capsule. Helps with evading immune destruction.

Symptoms: Prolonged fever (after 1 W - sustain up to 4 weeks), headache, and fatigue. Can be fatal. Splenomegaly, hepatomegaly if infection spread.

Big Picture

- Basic bacteriology
 - Classification (shape, stain, oxygen requirement)
 - Pathogenesis stages
 - Bacterial virulence factors.
 - Laboratory diagnosis
- Clinical Bacteriology
 - Clinically significant G-ve
 - Clinically significant G +ve

Resources

